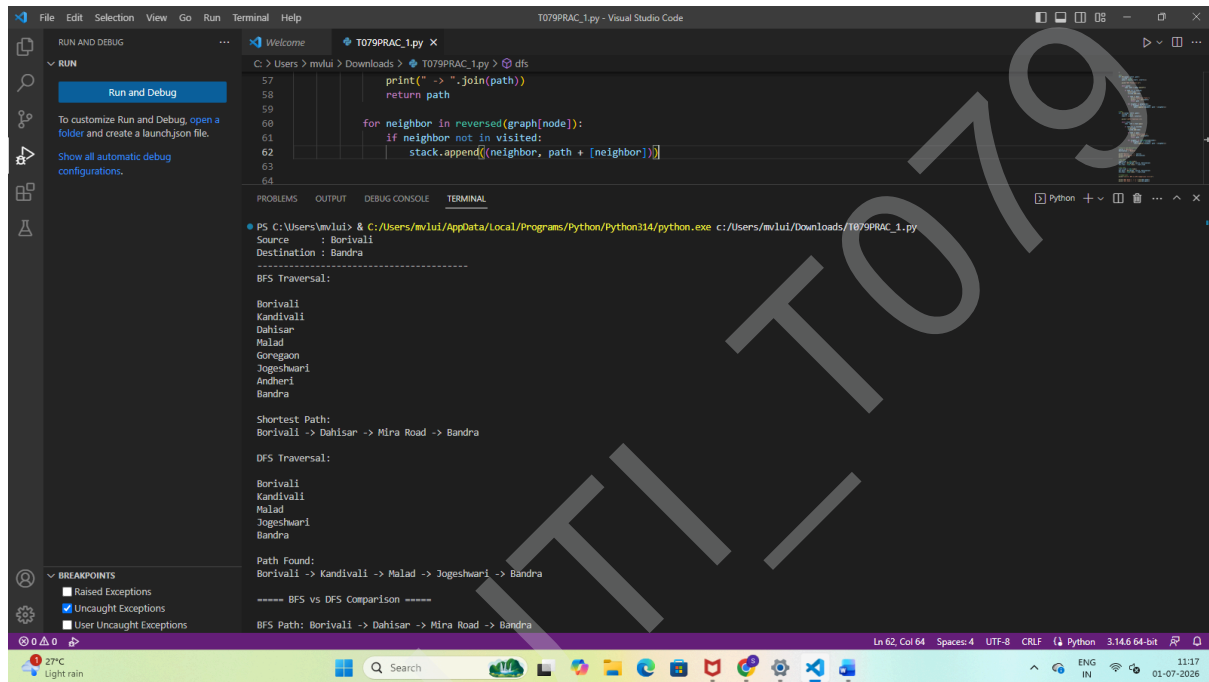


SHETH L.U.J. AND SIR M.V. COLLEGE

Practical 1

AIM:- Breadth First Search & Iterative Depth First Search in Python.

- Implement the Breadth First Search algorithm to solve a given problem.
- Implement the Iterative Depth First Search algorithm to solve the same problem.
- Compare the performance and efficiency of both algorithms.



The screenshot shows the Visual Studio Code interface with a Python file named T079PRAC_1.py. The code implements a Breadth First Search (BFS) algorithm to find the shortest path from Borivali to Bandra. The terminal output displays the BFS traversal path, the shortest path, and a comparison with DFS.

```
File Edit Selection View Go Run Terminal Help
T079PRAC_1.py - Visual Studio Code

RUN AND DEBUG
RUN
To customize Run and Debug, open a folder and create a launch.json file.
Show all automatic debug configurations.

T079PRAC_1.py X
C:\Users\mvlui> Downloads > T079PRAC_1.py > dfs
57 print("-> ".join(path))
58 return path
59
60 for neighbor in reversed(graph[node]):
61     if neighbor not in visited:
62         stack.append((neighbor, path + [neighbor]))
63
64

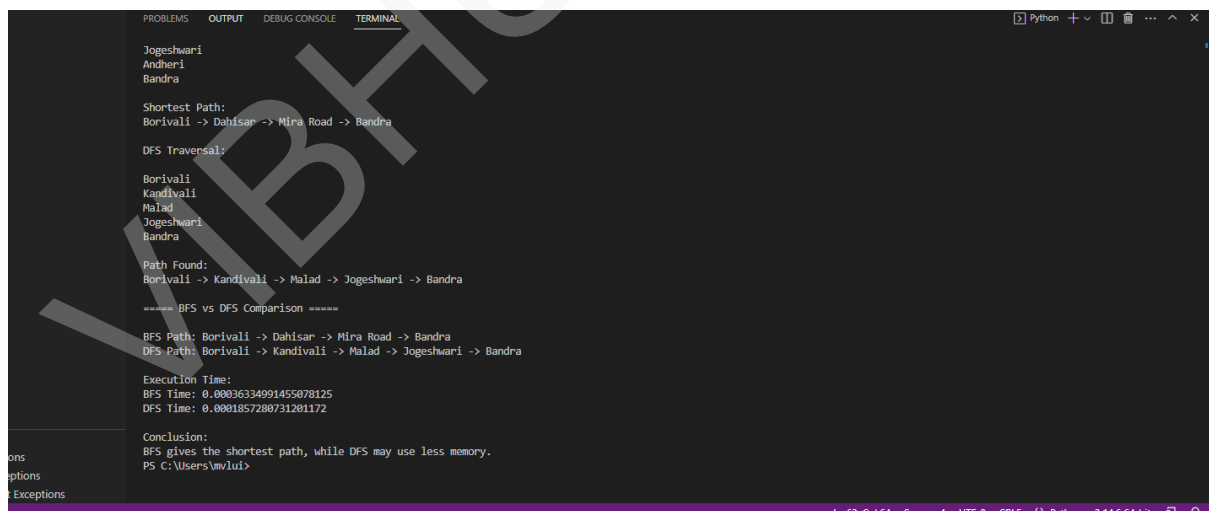
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\mvlui> & C:/Users/mvlui/AppData/Local/Programs/Python/Python314/python.exe c:/Users/mvlui/Downloads/T079PRAC_1.py
Source : Borivali
Destination : Bandra
-----
BFS Traversal:
Borivali
Kandivali
Dahisar
Malad
Goregaon
Jogeshwari
Andheri
Bandra

Shortest Path:
Borivali -> Dahisar -> Mira Road -> Bandra

DFS Traversal:
Borivali
Kandivali
Malad
Jogeshwari
Bandra

Path Found:
Borivali -> Kandivali -> Malad -> Jogeshwari -> Bandra

===== BFS vs DFS Comparison =====
BFS Path: Borivali -> Dahisar -> Mira Road -> Bandra
```



The continuation of the terminal output shows the DFS traversal, the path found, the comparison, and the execution time for both algorithms.

```

Jogeshwari
Andheri
Bandra

Shortest Path:
Borivali -> Dahisar -> Mira Road -> Bandra

DFS Traversal:
Borivali
Kandivali
Malad
Jogeshwari
Bandra

Path Found:
Borivali -> Kandivali -> Malad -> Jogeshwari -> Bandra

===== BFS vs DFS Comparison =====
BFS Path: Borivali -> Dahisar -> Mira Road -> Bandra
DFS Path: Borivali -> Kandivali -> Malad -> Jogeshwari -> Bandra

Execution Time:
BFS Time: 0.00036334991455078125
DFS Time: 0.0001857280731201172

Conclusion:
BFS gives the shortest path, while DFS may use less memory.
PS C:\Users\mvlui>
```